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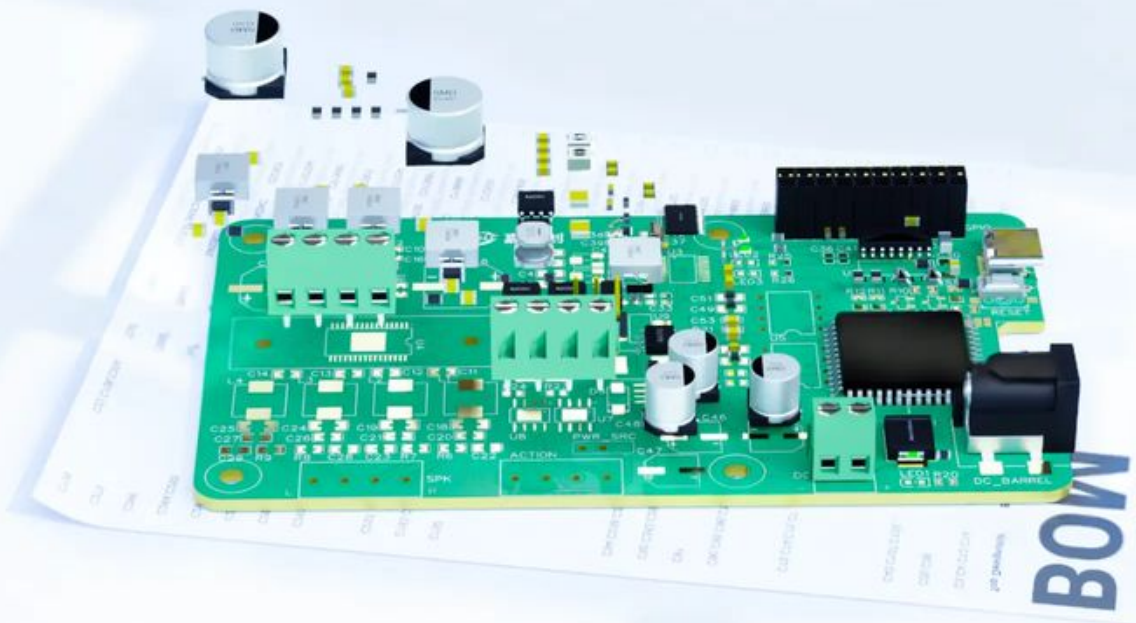
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# PCB Assembly Capabilities

Comprehensive PCB assembly capabilities for SMT, THT, and mixed-technology builds with rigorous inspection for prototypes, NPI, and volume production.

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Rigid PCB Capabilities | Flex PCB Capabilities | PCBA Capabilities | Stencil Capabilities | PCB Layout Capabilities | Flexible Heaters Capabilities

## PCB Assembly Technical Capabilities

| Features | Economic PCBA | Standard PCBA |
|---|---|---|
| Assembly Types | Single sided placement (SMT/Thru-hole) | Single & double sided placement (SMT/Thru-hole) |
| PCB Layer | 2,4,6-layers | 1 - 32 layers |
| Thickness | 0.8mm - 1.6mm | No limit |
| Dimension | Single PCB Size:<br>10×10mm - 470×500mm<br>PCB Panel Size:<br>10×10mm - 250×250mm | Single PCB Size:<br>70×70mm - 460×500mm<br>PCB Panel Size:<br>70×70mm - 250×250mm |
| Order Volume | 2 - 50 pcs | 2 - 80000 pcs |
| Surface Finish | Limited by specific options<br>(Refer to the options for Economic PCBA in the table below) | No limit |
| PCB Color | Limited by specific options<br>(Refer to the options for Economic PCBA in the table below) | No limit |
| Delivery Format | Single PCB, Panel with mouse bites | Single PCB, Panel with mouse bites, Panel with V-cut |
| Layer Stackup | Standard stack-up only, special stack-up is not supported | All stack-up |
| Gold Fingers/Castellated Holes/ Edge Plating | Not Support | Support |
| Edge Rails | Not necessary | Necessary |
| Fiducials | Not necessary | Necessary |
| Minimum Package | 0402 | 0201 |
| Minimum IC Pin Spacing | 0.4mm | 0.35mm |
| Minimum BGA Spacing | 0.5mm(center to center) | 0.3mm(center to center) |
| Reflow temperature | 255+/-5 °C (not adjustable) | 240+/-5 °C |
| SPI | No | Yes |
| AOI | Yes | Yes |

| | | |
|---|---|---|
| Visual Inspection | Yes | Yes |
| X-ray Inspection | Yes (only for certain parts, such as BGA) | Yes (only for certain parts, such as BGA) |
| Build time | 1 - 3 days | ≥ 4 days |

**PCB Specs for Economic PCB Assembly**

# FAQs about JLCPCB PCB Assembly Capabilities

**What are the minimum and maximum board sizes supported for PCBA?**

**Single PCB Assembly**

Standard PCBA: 70 × 70 mm to 460 × 500 mm

Economic PCBA: 10 × 10 mm to 470 × 500 mm

**Panelized PCB Assembly**

Standard PCBA: 70 × 70 mm to 250 × 250 mm

Economic PCBA: 10 × 10 mm to 250 × 250 mm

For panelized PCBs, larger panels are more susceptible to bending and deformation during SMT processing. Therefore, we recommend keeping panel dimensions within 250 × 250 mm whenever possible.

If your panel slightly exceeds the recommended size and the design requires it, production may still be feasible.

**Why is the maximum panel size for PCBA much smaller than the maximum single-board size?**

**Does JLCPCB charge extra tooling for flex PCB assembly, and are there any placement limitations around bend areas?**

**Do you support FPC (Flexible PCB) assembly?**

**What is the maximum PCB panel size for assembly?**

**Can you assemble THT components? Do you support mixed assembly (SMD + THT)?**

**What happens to unused through-holes during THT assembly? Will they be filled with solder?**

**Do you support fine-pitch ICs such as BGA and QFN?**

**Can JLCPCB manufacture and assemble 01005-size components?**

**Do you provide X-ray inspection for hidden solder joints?**

**Can you solder wires to a PCB for a power connection, like this? The wire is 22 AWG, 300 V rated, and 150 mm long. Can I create a part number for this wire?**

**Do you offer MCU programming? Can I send a .hex file to program the MCU before it is soldered onto the PCB?**

**Can JLCPCB assemble stacked components, such as a DIP IC inserted into a DIP socket, or a microcontroller mounted on a pin header/socket assembly?**

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